$(Bi_xSb_{1-x})_2Te_3$ Topological Insulators

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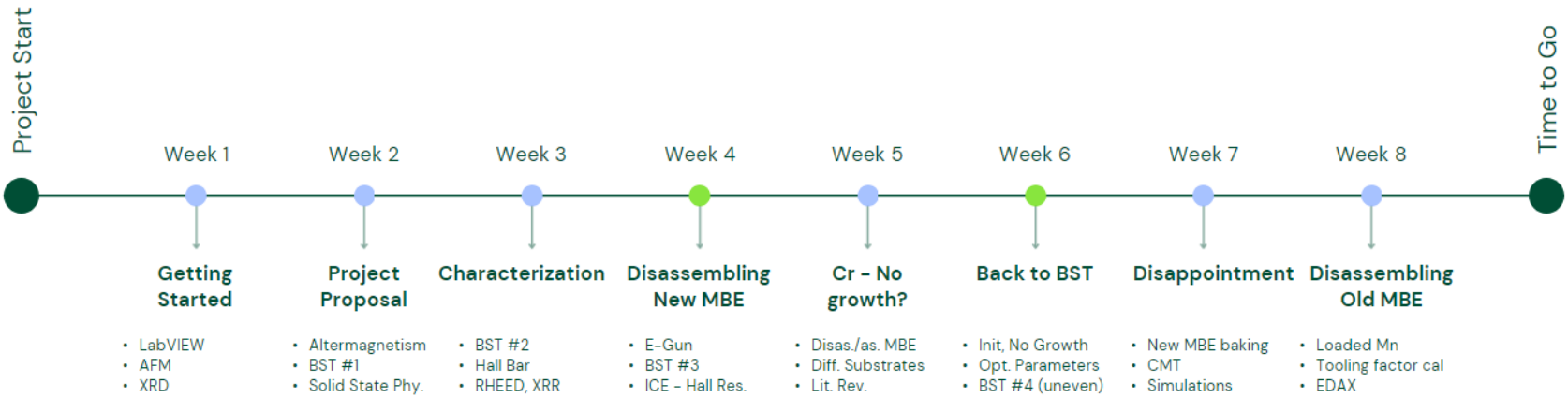
Summer Project Under the Guidance of Prof. **Karthik V. Raman**

Tata Institute of Fundamental Research, Hyderabad

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Timeline

Breakdown of My Visit



Introduction

What are Topological Insulators?

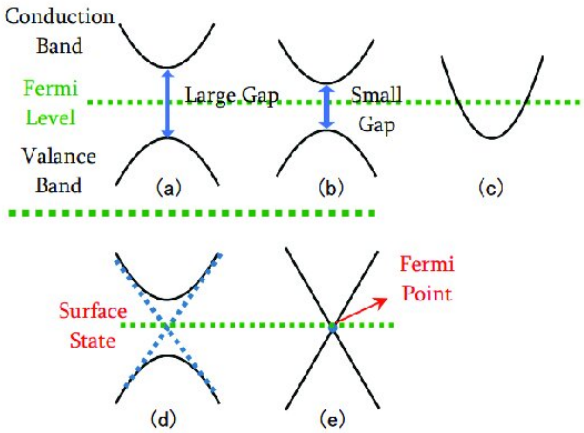


Figure: Band structures of different materials

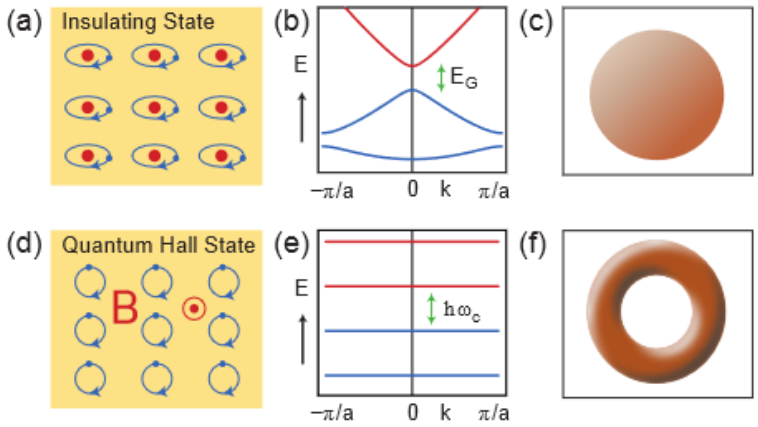


Figure: Insulating vs. quantum Hall states.

Topological insulators have insulating interiors but conductive surfaces or edges, with surface states protected by time-reversal symmetry, making them robust against impurities and defects.

- Fault-Tolerant Quantum Computing
- Thermoelelectric and Spintronic Devices

Bismuth Antimony Telluride

Motivation

1. Well known TIs such as Bi_2Te_3 and Sb_2Te_3 are extensively studied, so it's natural to consider their combinations to probe new class of materials.
2. **Better Control Over Gate Tun-ability**: Magnetically doped TIs such as Cr:BST offer access to the quantum anomalous Hall effect, which in its ideal form, could become a future resistance standard needing only a small permanent magnet to activate a quantized resistance value. [1]
3. TI films can be made superconducting, using the proximity effect. BST can be used to build superconducting qubits for topologically protected quantum computers. A recent study tries to explore this using Nb-BST-Nb arrays. [2]

[1] https://doi.org/10.48550/arXiv.2205.06077

[2] https://doi.org/10.1038/s42005-023-01288-9

Substrate

Why Sapphire?

Sapphire (hexagonal crystal structure, a = 4.76 Å, c = 12.99 Å) is commonly used to grow thin layers of topological insulating films, including BST because:

- It is commercially available.
- **High Thermal Stability**: Excellent thermal stability, which is crucial for the high-temperature processes involved in growing topological insulators.
- **Chemical Stability**: Sapphire is chemically inert, providing a stable environment for the growth.
- **Surface Quality**: Sapphire substrates offer a smooth and high-quality surface, essential for the epitaxial growth of topological insulator films, ensuring better uniformity and fewer defects.

Crystal Structure

- $(Bi_xSb_{1-x})_2Te_3$ compounds crystallize in a tetradymite-type structure, which is a rhombohedral crystal system often described using hexagonal lattice parameters for convenience. (a = 4.28 Å, c = 30.48 Å)
- The material forms a layered structure where quintuple layers (Te-Bi/Sb-Te-Bi/Sb-Te) are stacked along the c-axis with weak van der Waals interactions between them.

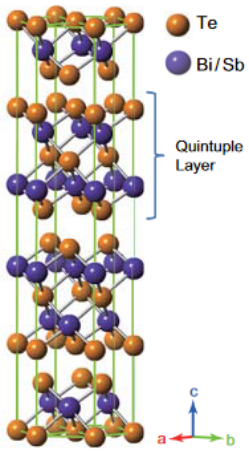


Figure: Crystal Structure of BST

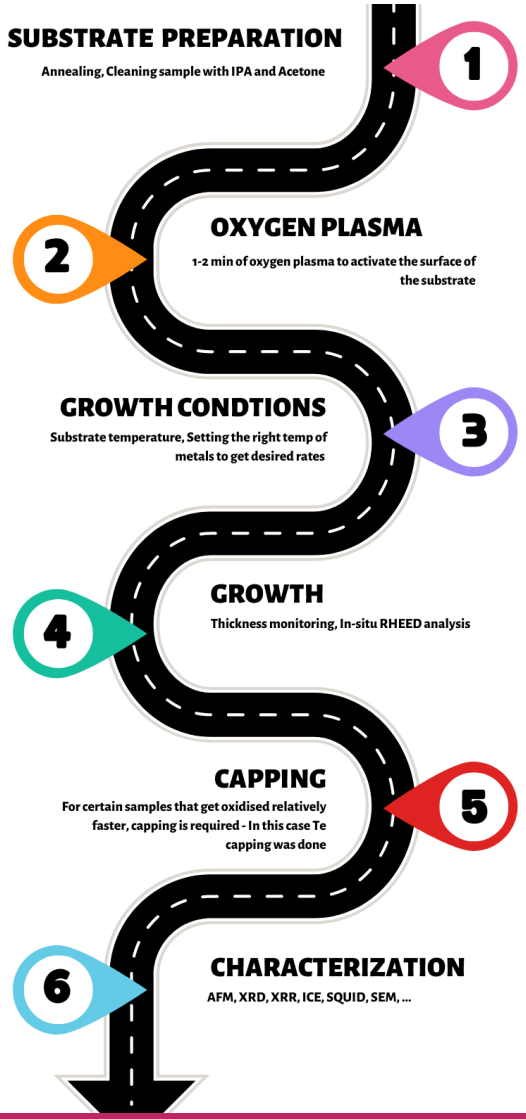
Lattice Mismatch with Sapphire

The lattice mismatch (Δa) between BST and sapphire can be estimated using the formula:

$$\Delta a = \frac{a_{sapphire} - a_{BST}}{a_{sapphire}} \times 100\%$$

Plugging in the values, we get $\Delta a \approx 10.08\%$ which is significant but can be managed through appropriate growth techniques to ensure high-quality epitaxial films.

A Journey to Growing Samples



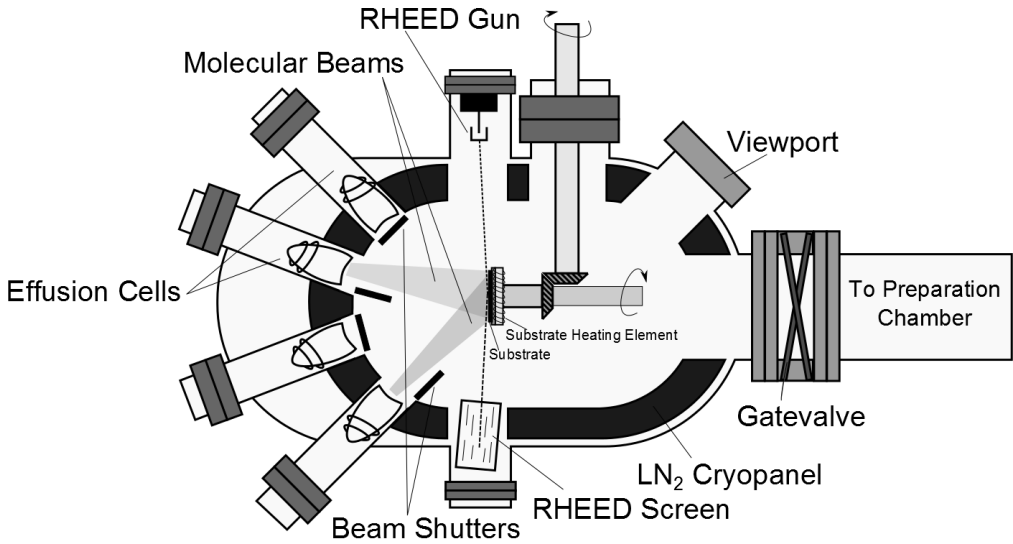


Figure: Molecular Beam Epitaxy

Sapphire Substrate

Annealing, Terraces and O_2 Plasma

A 5mm $\times$ 5mm sapphire substrate was annealed at 1300 $^{\circ}C$ and was treated with O_2 plasma for 50 sec.

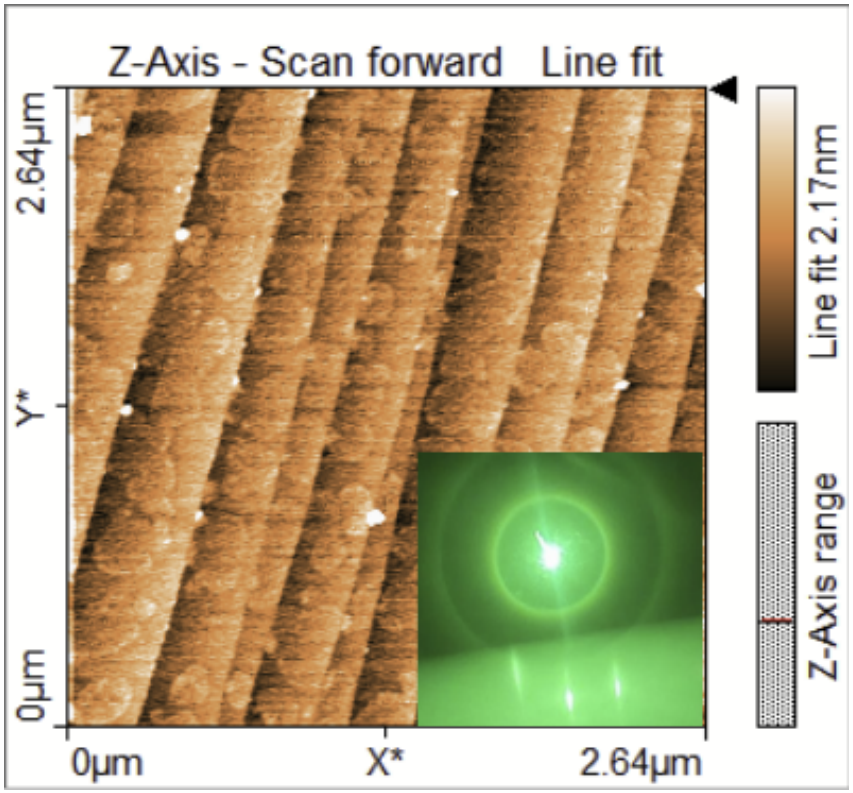


Figure: Surface Morphology of Annealed Sapphire with In-situ RHEED

BST #1 and #2

The samples were grown at substrate temperatures 180°C for 1 hr and 200°C for 45 min respectively. Target x = 0.2, Bi (0.04 Ås^{-1}, 555°C) / Sb (0.1 Ås^{-1}, 503°C) / Te (1.25 Ås^{-1}, 426°C)

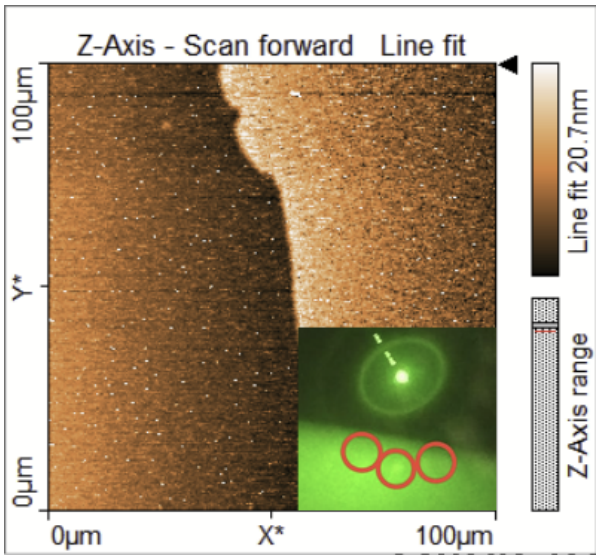


Figure: Surface Morphology of BST #1 with In-situ RHEED. The red circles mark the diffraction spots. Thickness = 18.96nm

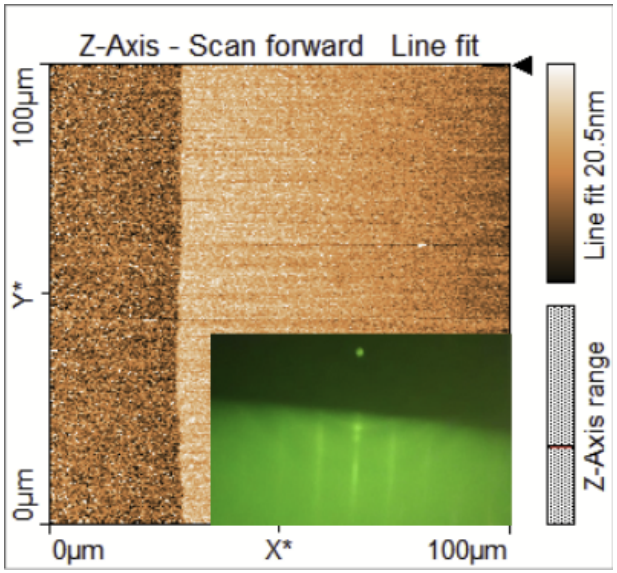


Figure: Surface Morphology of BST #2 with In-situ RHEED. Thickness ≈ 9-10nm

BST #3

1. The sample was grown at 220°C substrate temperature for target x = 0.2 for 27 minutes. It was then annealed post growth at 280°C for 30 min and was capped with a layer of Te for about 7 min (≈7nm thickness.)
2. A typical six contact hall bar geometry was scratched onto the surface of the film by hand, which was then wire bonded and put into the cryostat for MR measurement.

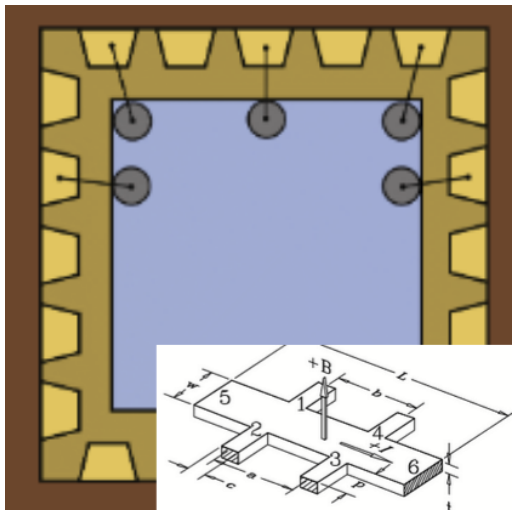


Figure: Wire-bonding and hall bar geometry

Hall Measurement

Sample #3

The R vs. T graph shows a decreasing resistance with increasing temperature, suggesting non-topological behavior typical of a bulk conductor, where phonon scattering decreases the resistance, unlike the expected plateau or slight increase for a topological insulator due to protected surface states. The carrier density calculated was negative and of the order $10^{-25}\ m^{-3}$ suggesting bulk conductivity.

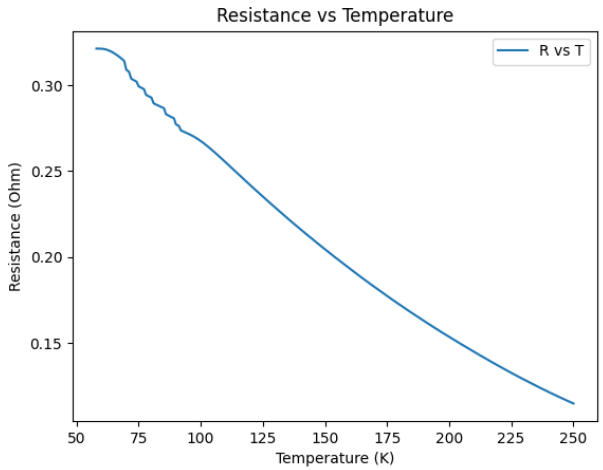


Figure: Resistance v/s Temperature

EDAX: Energy-dispersive X-ray spectroscopy

Sample #4

Target x = 0.4, Bi (0.1 $Å s^{-1}$) / Sb (0.2 $Å s^{-1}$) / Te (1.25 $Å s^{-1}$)

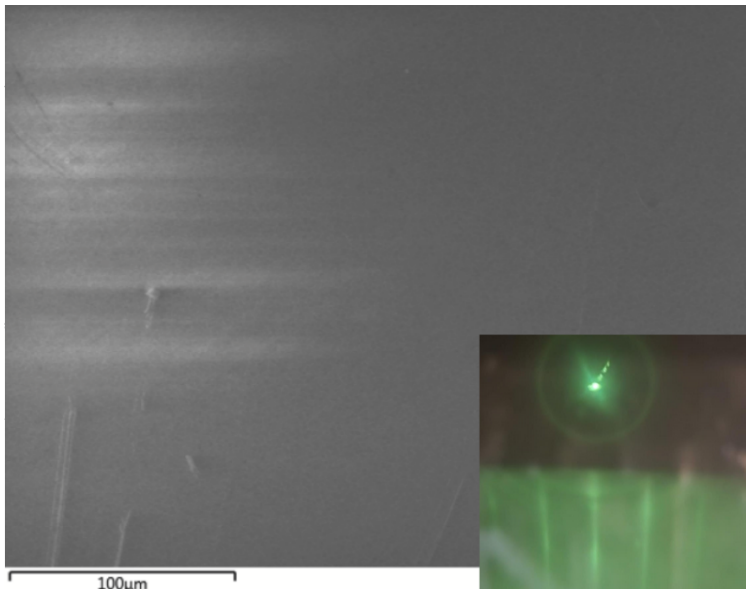


Figure: Electron Image and In situ RHEED post annealing at 250 $°C$

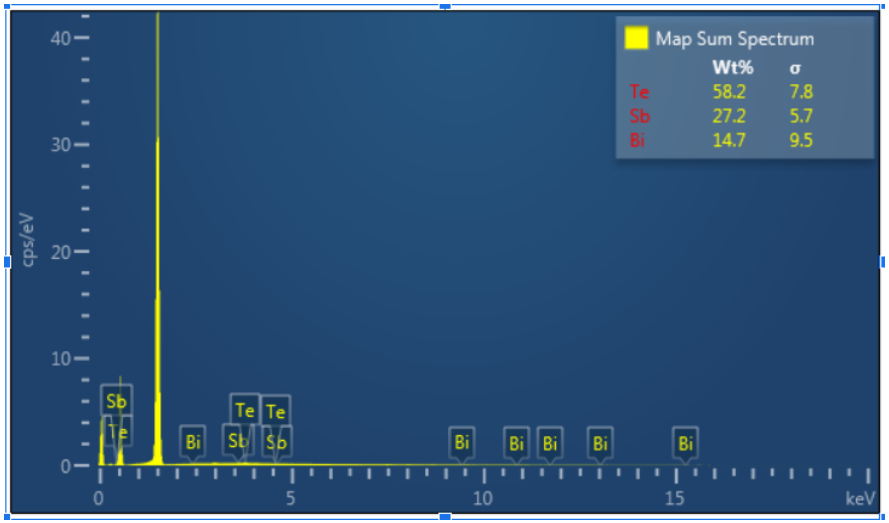


Figure: EDAX analysis I

EDAX: Energy-dispersive X-ray spectroscopy

Composition Analysis

Atomic Weights:
Te: 127.60 g/mol, Sb: 121.76 g/mol, Bi: 208.98 g/mol

Moles Calculation:
$n_{Te} = \frac{58.2}{127.60} = 0.4561$, $n_{Sb} = 0.2234$, $n_{Bi} = 0.0704$

Normalized Moles:
$\text{Te} = \frac{0.4561}{0.0704} = 6.48$, $\text{Sb} = \frac{0.2234}{0.0704} = 3.17$, $\text{Bi} = \frac{0.0704}{0.0704} = 1$

Composition: $x = \frac{1}{1+3.17} \times 2 = \frac{1}{4.17} \times 2 \approx 0.48$

Calculated Composition: $(\text{Bi}_{0.48}\text{Sb}_{0.52})_2\text{Te}_3$

EDAX: Energy-dispersive X-ray spectroscopy

Sample #4

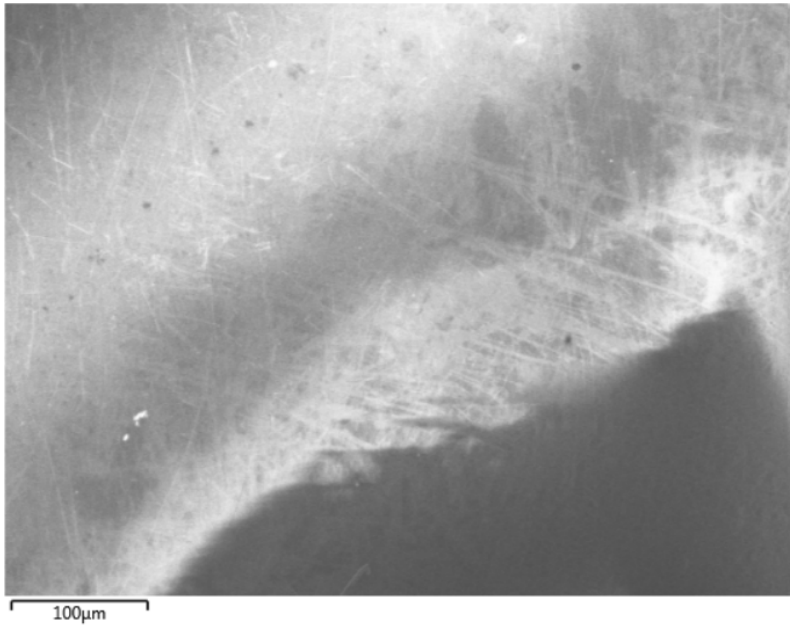


Figure: Electron Image at a different region

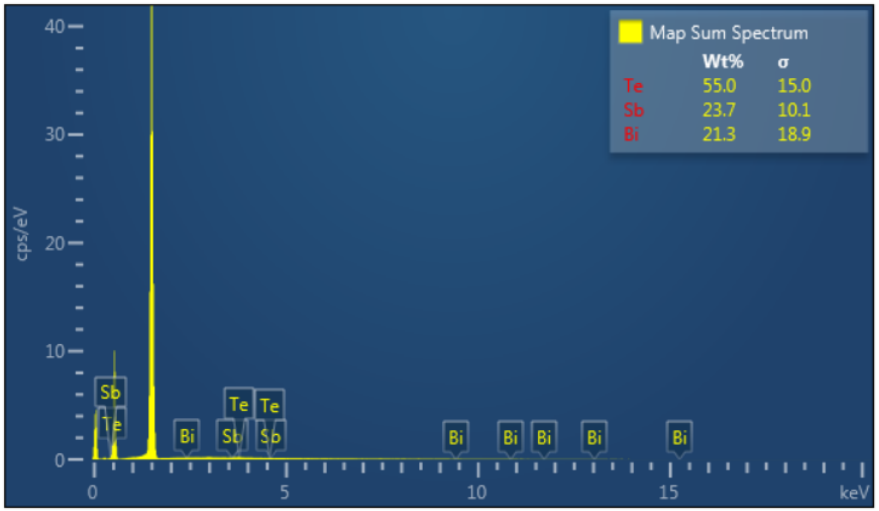


Figure: EDAX analysis II

Calculated Composition: $(Bi_{0.69}Sb_{0.31})_2Te_3$

Issues

Challenges faced during the process

1. High fluctuations in the rate of tellurium (0.9 - 1.5 Å/s) during the growth affected the quality of the films.
2. The thickness monitors weren't rightly calibrated for the elements, hence the actual thickness drastically varied from the expected one.
3. High Te flux was responsible for etching off the film causing uneven growth.
4. The cryostat results were irregular and the values generated were much higher than the usual ones.

Way to Move Forward

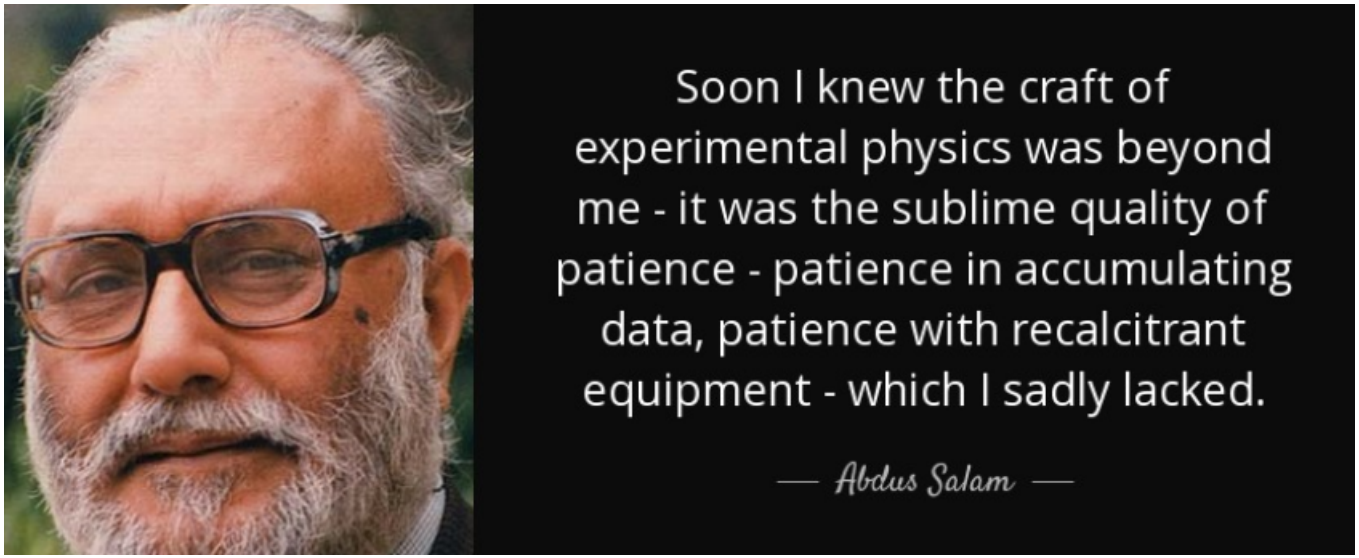
Immediate Goals

1. Growing various thin films of BST with the dopant x $\in (0, 1)$ and plotting the change in carrier density as a function of doping to find the ideal composition.

2. Start looking at Cr:BST when Cr-rod arrives and reproduce some of the existing literature results.

3. Interfacial enigneering and studying proximity effects to gain insights towards realizing logical qubits.

Key Learnings



- As easy as it is to come up with ideas, it is just as difficult to implement them.
- Focus on finding ways to work with limited resources instead of waiting for things to change.

Project Proposals

I would like to see get implemented

- To study the 2D topological altermagnetic behaviour in thin films of CrSb.
  (https://doi.org/10.48550/arXiv.2405.14777)
- To experimentally verify the contributions of the Quantum Metric Dipole in the Non-Linear Transport of CrSe.
  (https://doi.org/10.48550/arXiv.2406.03738)
- Suprisingly, there is no experimental literature on CrBi at all, so to reproduce the room tempertaure ising behaviour and tunable magneto-optical response from theoretical predictions.
  (10.1103/PhysRevApplied.15.064053)
- To experimentally probe the superconductivity of ultra-thin films of Au, Ag and Cu.
  (https://doi.org/10.48550/arXiv.2406.16621)

Conclusion

Epitaxial thin films of BST of about 8 - 10 nm were fabricated using the Molecular Beam Expitaxy system, which were then further characterized using AFM, RHEED, XRD, ICE and EDAX to study its magneto-resistance and structural properties to draw inferences about its topological character.

Skills Acquired:

- NanoSurf, Gwyddion (AFM)
- Rigaku SmartLab (XRD, XRR)
- LabVIEW (Automation)
- Molecular Beam Epitaxy (Thin Films)
- Scratching, Wire-Bonding (ICE)

Acknowledgments

I would like to thank Prof. Kathik V. Raman for providing me with this opportunity to learn under his guidance. I would also like to thank Mr. Swapnil for his help in growing the BST films and in performing AFM and XRD, Dr. Suman for providing the Hall Bar Measurements and Dr. Malvika for providing the EDAX results. Lastly, I would like to thank Prof. Ajaya Kumar Nayak for suggesting that I reach out to the Raman Lab for my summer project.